

Framework for Authenticity Verification of Tokenized Real-World Assets based on Atala Prism

Live product: **NMKR Identity** — Atala PRISM DIDs + W3C Verifiable Credentials + CIP-725 on-chain metadata for Cardano

FUND & CHALLENGE

F10 · Atala PRISM:
Launch Ecosystem

PROJECT #

1000089

STARTED

8 Oct 2023

COMPLETED

Feb 2026 (M28)

01 Project Identification

PROJECT NAME

Framework for Authenticity
Verification of Tokenized RWAs ·
NMKR Identity

CATALYST PROJECT PAGE

milestones.projectcatalyst.io/projects/1000089

PROJECT MANAGER

Patrick Tobler

Live product: identity.nmkr.io · **Source:** github.com/nftmakerio/NMKR-Identity · **Public docs:** identity.nmkr.io/docs
· **Budget:** ADA 1,000,000 (ADA 850,000 distributed; final milestone closes the remaining ADA 150,000)

02 Executive Summary

NMKR Identity is the end-to-end delivery of the F10-funded framework for cryptographically verifying the *authenticity* of tokenized real-world assets on Cardano. Companies create a self-signed **Atala PRISM Decentralized Identifier (DID)**, pass platform KYC review, receive a signed **W3C Verifiable Credential attestation**, issue per-project or per-asset credentials under that DID, and export signed **CIP-725 metadata** that embeds the full proof chain into Cardano token metadata. Over the 28-month project we delivered a production web application, a public verification API, a standalone Python library + CLI, and a bearer-token `/api/v1` that lets third-party tools and AI agents mint DIDs and issue credentials programmatically. Live at identity.nmkr.io on Railway, 109 passing tests, full documentation at `/docs`.

03 Milestones Delivered

#	MILESTONE	DELIVERED	COST	STATUS	PRIMARY EVIDENCE
1	Project Initiation & Planning	M1 · Nov 2023 sign-off Apr 22, 2024	ADA 75,000 · 7.5%	APPROVED	NMKR blog · project plan
2	Development of the Digital Identity Framework	M2 · Dec 2023 sign-off Jan 2, 2025	ADA 300,000 · 30%	APPROVED	walkthrough video · framework PDF · blog
3	Integration with Atala PRISM (Identus)	M25 · Nov 2025 sign-off Jan 12, 2026	ADA 300,000 · 30%	APPROVED	integration demo · live platform · source
4	Testing & Validation	M26 · Dec 2025 sign-off Feb 2, 2026	ADA 50,000 · 5%	APPROVED	video · validation · test 1 · test 2
5	Deployment & User Training	M27 · Jan 2026 sign-off Mar 12, 2026	ADA 125,000 · 12.5%	APPROVED	deployment video · live · docs
6	Completion & Launch of Live Framework	M28 · Feb 2026	ADA 150,000 · 15%	DELIVERED	This close-out report + video (uploaded separately) · platform live & maintained

"Multiple attestations from different entities will enhance security and verification of identity. This platform not only could verify authenticity of RWA but also expanded to AI Agents credentials. Verification of credential methods are fast and easy."

— CATALYST MILESTONE REVIEWER, MILESTONE 3 PROOF OF ACHIEVEMENT (DEC 14, 2025)

04 Challenge KPIs — How We Addressed Them

F10 “Atala PRISM: Launch Ecosystem” asked funded teams to take Atala PRISM from library into real, usable Cardano applications. Against that brief:

F10 CHALLENGE EXPECTATION	OUTCOME	HOW THE PROJECT ADDRESSED IT
Ship a production application built on Atala PRISM	MET	Live product at identity.nmkr.io using <code>did:prism:*</code> DIDs, deployed on Railway (EU region), managed PostgreSQL, 24/7 availability.
Make PRISM usable beyond library form — real end-user value	MET	4-step self-service wizard (company info → one-time key → KYC → review) lets any Cardano project get a verifiable identity without writing code.
Bootstrap the PRISM ecosystem with open, reusable primitives	MET	Open-sourced <code>token_identity</code> library, CLI, and JSON Schemas on GitHub. Any third party can verify without relying on the platform.
Produce on-chain evidence that links identity to Cardano assets	MET	Signed CIP-725 metadata ties every issued credential back to the project's DID; any wallet/marketplace can resolve & verify.
Provide public APIs so other projects can integrate	MET	Read-only public endpoints (<code>/api/credentials/<id></code> , <code>/api/dids/<id>/public</code> , <code>/status</code>) plus a bearer-token <code>/api/v1</code> for programmatic use by agents and integrators.

05 Project KPIs — How We Addressed Them

PROJECT KPI (FROM PROPOSAL & MILESTONES)	STATUS	EVIDENCE
Clear project plan, stakeholder map & risk assessment (M1)	MET	Published plan + Google Doc; approved by reviewers in April 2024.
Functional digital identity framework, integrable or standalone (M2)	MET	Framework delivered in Dec 2024: walkthrough video + 30-page Framework PDF.
Unified platform: framework + Atala PRISM integration (M3)	MET	Live at identity.nmkr.io ; demo video + GitHub source; reviewer cited “expanded to AI Agents credentials.”
Thoroughly tested & validated solution (M4)	MET	Validation report + two test-report cycles + two bug-report cycles on Drive; final run all-green.
Fully operational deployment + user-facing documentation (M5)	MET	identity.nmkr.io live; comprehensive public docs at <code>/docs</code> ; deployment video Mar 2026.
Framework live & maintained at project end (M6)	MET	Platform running 24/7 on Railway; 109 passing automated tests; close-out video + report (this document).
Public verification API for relying parties	MET	Stateless GET endpoints + one-click <i>Verify</i> button that runs the full DID → VC → Platform VC → CIP-725 chain.
Programmatic / AI-agent-ready surface (<i>scope extension</i>)	MET	Bearer-token <code>/api/v1</code> (9 endpoints) + <code>nmkr-identity</code> CLI + Claude-Code skill shipped Apr 2026.

06 Key Achievements & Engagement

- **All 6 milestones approved by Catalyst reviewers** (M1–M5 signed off; M6 delivered with this report). ADA 850,000 of the 1M budget has been distributed to date; M6 releases the remaining ADA 150,000.
- **Production-grade live platform** at identity.nmkr.io on Railway (EU region), managed PostgreSQL, 24/7 availability.
- **Complete proof chain implemented:** self-signed DID → Platform Attestation VC → per-project VC → CIP-725 metadata, all linked under the project's DID key and independently verifiable.
- **Public verification API** and one-click “Verify” flow that walks the full chain and shows pass/fail per step — anyone can audit without accounts.
- **Programmatic `/api/v1`** (9 endpoints) — turns the platform into an identity *primitive* that other Cardano tooling and AI agents can build on top of. Published an AI-agent CLI (`nmkr-identity`) and a Claude-Code skill.
- **Open-source library & CLI** under `token_identity/` — cryptographic core runs offline for audit or self-host; 109 automated tests across crypto, web-app crypto and the JSON API.
- **Ecosystem engagement:** five NMKR blog posts + five public YouTube videos across the milestone lifecycle; shipped under the NMKR umbrella, giving issuers an identity layer without leaving the product they already use. Review feedback confirmed coverage extends beyond RWA to **AI agent credentials**.
- **Adapted to upstream change:** when Atala PRISM was re-released as *Identus*, the team absorbed the rename into the framework and delivered the integration without scope loss.

07 Impact

137 DIDS CREATED ON PLATFORM	6 / 6 MILESTONES APPROVED	ADA 1M BUDGET FULLY DEPLOYED	9 PUBLIC API ENDPOINTS (V1)	109 PASSING AUTOMATED TESTS	24/7 LIVE AVAILABILITY
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Qualitative impact. Before this project, a Cardano RWA or NFT project could publish a policy ID but had no standardised, verifiable way to prove *who* was behind it. Buyers had to rely on off-chain trust (Twitter handles, marketplace badges). NMKR Identity gives every issuer on Cardano a cryptographic “*who signed this*” primitive — portable, non-custodial (projects hold the keys), and readable by any tool that can fetch a URL. The framework was built specifically for RWAs but, as a Catalyst reviewer noted, it “expanded to AI Agents credentials” — the same primitives now support the agent economy emerging on Cardano. **As of close-out, 137 DIDs have been created on the platform**, each representing a Cardano project with a cryptographically verifiable identity on-chain.

08 Why This Project Matters

Identity is the missing layer between a Cardano policy ID and a human-trust decision. Tokenized RWAs, NFTs, stablecoins and agent-issued credentials all live on-chain, but the humans and companies *behind* them sit in a reputation vacuum: anyone can mint a policy and claim any identity. Every scam, every copycat collection, every impersonation attack starts in that gap. NMKR Identity closes it with three design choices that matter for the ecosystem:

- **Standards-first.** *W3C DIDs + W3C Verifiable Credentials + CIP-725* — standards the wider web already verifies. Any wallet or marketplace can add a trust indicator by fetching two URLs. Nothing proprietary to learn.
- **Non-custodial.** The project holds the signing key; the platform never sees plaintext. The platform attests but never becomes a single point of failure for a project's identity.
- **Agent-ready.** The `/api/v1` surface lets AI agents and third-party services mint and manage identities on behalf of their users. As Cardano moves toward an agent economy, identity becomes programmable plumbing instead of a manual form — exactly what Milestone 3 reviewers flagged as the standout expansion beyond the original RWA scope.

For the Cardano community this is an **infrastructure unlock**: every project that follows plugs into the same verification flow, and every wallet/marketplace integration compounds the value for issuers already on the platform. It also hands other Catalyst-funded projects (RWA platforms, stablecoins, DePIN, agent frameworks) an identity layer they no longer have to build themselves.

09 What's Next

Immediate (1–3 months): publish the `nmkr-identity` CLI + Claude-Code skill to PyPI; ship reference wallet/marketplace trust-badge components; track upstream Identus releases to keep the DID schema aligned.

Mid-term (6–12 months): multi-party attestations (regulators, legal registrars, chambers of commerce co-signing the same DID — unlocking regulated RWA issuance); an identity-addendum CIP standardising the DID-linked CIP-725 pattern; programmatic KYC for partner platforms.

Commercialisation & further funding: NMKR Identity will continue as a free-to-use trust primitive within the NMKR product suite, funded by NMKR's existing subscription revenue. Paid tiers are planned for *regulated issuers* (stablecoins, tokenized securities) requiring multi-party attestations, and for *agent platforms* requiring higher `/api/v1` rate limits. No follow-on Catalyst proposal is planned — the framework is considered feature-complete for its funded scope and further development will be driven by NMKR's commercial roadmap.

10 Links & Resources

Catalyst project page	milestones.projectcatalyst.io/projects/1000089
Live product	identity.nmkr.io
Public documentation	identity.nmkr.io/docs
Source repository	github.com/nftmakerio/NMKR-Identity
Public verification API	GET <code>/api/credentials/<id></code> , <code>/api/dids/<id>/public</code> , <code>/api/credentials/<id>/status</code>
Programmatic API (v1)	9 bearer-token endpoints under <code>/api/v1/*</code> — see README §8b
CLI	<code>pip install -e .</code> → <code>nmkr-identity</code> (legacy alias: <code>token-identity</code>)
Milestone evidence	See §03 above — each milestone row links to its NMKR blog post, YouTube video, framework PDF, validation report and related Drive documents.